

HTTP (HyperText Transfer Protocol)

Core Characteristics

Text-Based

- HTTP messages are plain text.
- Requests and responses consist of **headers** and an **optional body**.
- This makes HTTP easy to inspect and debug (browser dev tools, curl, Postman).

Client–Server Protocol

- HTTP follows a strict client–server architecture.

Clients Initiate All Communication

1. The **client** (browser, app, Postman) always sends the first message.
2. The **server never initiates** communication.
3. The server only **responds** to client requests.
4. This asymmetry defines the client–server model.

Stateless

- Each request is independent.
- The server does **not remember** prior requests.
- State must be added explicitly (cookies, sessions, JWTs).

Two Types of Messages

1. **Request**
 - Headers
 - Optional body
2. **Response**
 - Headers
 - Optional body

HTTP Status Code Categories

100-level — Informational

- Request received, processing continues.
- Example:
 - **100 Continue**

200-level — Success

- Request succeeded.

- Example:
 - 200 OK
 - 201 Created

300-level — Redirection

- Client must take additional action (usually another request).
 - Example:
 - 301 Moved Permanently
 - 302 Found
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Headers vs Body

Headers

- Metadata about the request or response.
- Examples:
 - Content-Type
 - Content-Length
 - Host
 - User-Agent

Body

- Actual payload/data.
- Examples:
 - HTML
 - JSON
 - File data
- **Optional** — not all requests or responses include a body.

Is There Always a Body in a Response?

- **No.**
- Examples with no body:
 - 204 No Content
 - Responses to HEAD requests
 - Some redirects

Example body placeholder: